

YEAR 7 MATHEMATICS

Chapter 1

Integers

LEARN IT · PRACTISE IT

1

How to use this booklet

1 **Page 1: learn it.** Start with the **Do now**, from memory. Then read each worked **Example** and do the **Your turn** beside it. Each **Your turn** gives you a little less help.

2 **Check.** Everyone answers the one question at the end of page 1. It tells your teacher who is ready for page 2.

PAGE 2 **Spot the pattern, then Mixed practice:** today's skill mixed with earlier ones, so you have to choose the method. Finished? **Go further.**

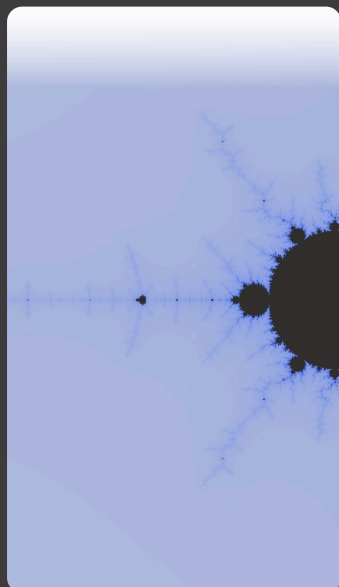
EXIT The **Exit ticket** is on your own. Circle how you went: Not yet, Nearly or Got it.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
1.01	Integers and opposites	3		☐
1.02	The number line	5		☐
1.03	Comparing and ordering integers	7		☐
1.04	Adding integers	9		☐
1.05	Subtracting integers	11		☐
1.06	Multiplying integers	13		☐
1.07	Dividing integers	15		☐
1.08	Order of operations with integers	17		☐
1.09	Integers on a calculator	19		☐
1.10	Integer problems	21		☐
Review	Chapter review	23		☐

In this chapter you will

- understand positive and negative numbers and their opposites
- use a number line to show and compare integers
- order integers
- add, subtract, multiply and divide integers
- use the order of operations with integers, with and without a calculator
- solve problems involving integers



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \rightarrow z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is the long antenna of the set along the negative real axis, dotted with tiny copies of the whole set. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

1.01 Integers and opposites

YEAR 7 · CHAPTER 1: INTEGERS

NAME _____ CLASS _____ DATE _____

DO NOW From memory. No notes.

1 Order smallest to largest: 12, 3, 30, 0

2 What is $8 - 8$?

3 It is 0°C . It gets 4 degrees colder. What is the temperature?

4 What number is halfway between 0 and 10?

THINK FIRST Have a go before we learn it. It's fine to be unsure.

A lift is at the ground floor, level 0. It goes down 2 floors to the car park. Which level is that?

WORDS

integer: a whole number: ... -2, -1, 0, 1, 2 ...

opposite: same distance from 0, other side.

$-(-5)$: say "the opposite of negative five".

BIG IDEA

Negative numbers are left of 0. A number and its opposite are the same distance from 0, on opposite sides.



LEARN IT Read each example. Then do the one beside it. Stuck? Look back at the example.

EXAMPLE 1

Write an integer for 5 m below sea level.

below 0: negative

= -5

Why? Why is "below sea level" negative?

YOUR TURN 1

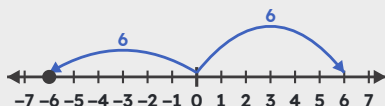
Write an integer for a loss of \$8.

a loss: _____

= ____

EXAMPLE 2

The opposite of -6



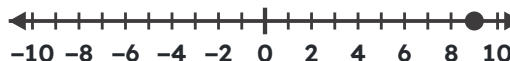
-6 is 6 steps left of 0

opposite: 6

Why? What is the opposite of 0? Why?

YOUR TURN 2

The opposite of 9



___ is ___ steps right of ___

EXAMPLE 3

$-(-7)$

= the opposite of -7

= 7

Why? Read $-(-7)$ aloud, using the word "opposite".

YOUR TURN 3

$-(-11)$

CHECK Everyone answers. Circle one letter, then hold it up.

What is the opposite of -8?

- (A) -8 (B) 8 (C) 0 (D) -7

SPOT THE PATTERN

Find each one. Each question changes one thing.

1 opposite of 3 = _____

2 opposite of 2 = _____

3 opposite of 1 = _____

4 opposite of 0 = _____

5 opposite of -1 = _____

6 opposite of -2 = _____

7 $-(-2) =$ _____

8 $-(-20) =$ _____

What do you notice? The number goes down by 1 each time. What happens to its opposite?

MIXED PRACTICE

Today's skill mixed with earlier ones. Read each question carefully: they are not all the same kind.

1 Write an integer for 12°C below zero.

1.01

2 What is the opposite of 15?

1.01

3 $-(-9)$

1.01

4 Which is further from 0: -7 or 5?

1.01

5 A diver is 20 m below sea level. Write this as an integer.

1.01

6 List the integers between -3 and 2.

Year 6

7 The opposite of the opposite of 6

1.01

8 Write an integer for a gain of 7 points.

1.01

9 What is the opposite of -1?

1.01

10 Which two integers are 10 steps from 0?

1.01

GO FURTHER

Finished? Try these. Explain your thinking.

1 Mia says "the opposite of a number is always negative". Is she right? Explain.

2 Two opposites are 18 steps apart. What are they?

3 Which integers from -5 to 5 are more than 3 steps from 0?

EXIT TICKET

On your own. Hand it in.

1 Write an integer for 4°C below zero.

2 What is the opposite of -13?

3 $-(-2)$

Today I can use integers and find opposites.

Not yet

Nearly

Got it

1.02 The number line

YEAR 7 · CHAPTER 1: INTEGERS

NAME _____ CLASS _____ DATE _____

DO NOW From memory. No notes.

- 1 What is the opposite of -4 ? 1.01 2 $-(-10)$ 1.01 3 Write an integer for 3 floors below ground. 1.01 4 Count by 5s: 15, 20, 25, ... What comes next? Year 6

THINK FIRST Have a go before we learn it. It's fine to be unsure.

A dot is halfway between -4 and 0 . What number is it at?

WORDS

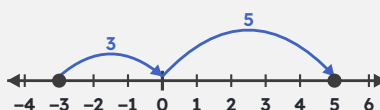
scale: how much each gap is worth.

distance: how many steps apart. Never negative.

halfway: the same distance from both ends.

BIG IDEA

Numbers get larger to the right. To find a distance, count the gaps. Across 0? Count to 0, then on from 0.

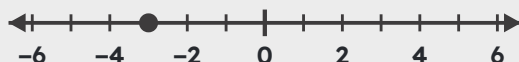


LEARN IT

Read each example. Then do the one beside it. Stuck? Look back at the example.

EXAMPLE 1

Which number is at the dot?



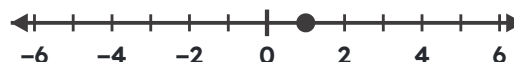
one gap left of -2

$= -3$

Why? Only every second mark has a number. How did we still find it?

YOUR TURN 1

Which number is at the dot?

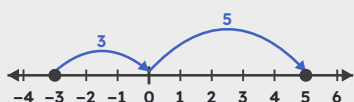


one gap right of ____

$=$ ____

EXAMPLE 2

How far is it from -3 to 5 ?



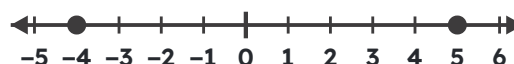
3 to 0, 5 from 0

$3 + 5 = 8$

Why? Why do we add the two parts?

YOUR TURN 2

How far is it from -4 to 5 ?



____ to ____, ____ from ____

EXAMPLE 3

Which integer is halfway between -8 and 2 ?

10 apart, half is 5

5 right of -8 : -3

Why? How can you check that -3 is halfway?

YOUR TURN 3

Which integer is halfway between -6 and 4 ?

CHECK Everyone answers. Circle one letter, then hold it up.

How far apart are -4 and 3 ?

- (A) 1

(B) 7

(C) -7

(D) -1

SPOT THE PATTERN

How far apart are the two numbers? Each question moves one end.

1 -1 to $3 =$ _____

2 -2 to $3 =$ _____

3 -3 to $3 =$ _____

4 -4 to $3 =$ _____

5 -4 to $2 =$ _____

6 -4 to $1 =$ _____

7 -4 to $0 =$ _____

8 -4 to $-1 =$ _____

What do you notice? What happens to the distance when one end moves 1 step outwards? 1 step inwards?

MIXED PRACTICE

Today's skill mixed with earlier ones. Read each question carefully: they are not all the same kind.

1 How far is it from -9 to -4 ? 1.02

2 What is the opposite of 7? 1.01

3 Start at -2 and move 7 right. Where do you land? 1.02

4 Which integer is halfway between -10 and 0? 1.02

5 $-(-3)$ 1.01

6 A number line goes $-20, -10, 0, \dots$ What are the next two numbers? 1.02

7 How far is it from -12 to 8? 1.02

8 Start at 4 and move 9 left. Where do you land? 1.02

9 Which is further from 0: -6 or 4? 1.01

10 A number line runs from -15 to 15. What is in the middle? 1.02

GO FURTHER

Finished? Try these. Explain your thinking.

1 Two integers are 12 apart. Halfway between them is -2 . What are they?

2 Start at -5 . Move 3 right, then 8 left, then 6 right. Where do you land? How many steps did you walk?

3 How many integers are between -6 and 4, not counting -6 and 4?

EXIT TICKET

On your own. Hand it in.

1 How far is it from -7 to 2?

2 Which integer is halfway between -10 and 4?

3 Start at 3 and move 8 left. Where do you land?

Today I can read number lines and find distances between integers.

Not yet

Nearly

Got it

1.03 Comparing and ordering integers

YEAR 7 · CHAPTER 1: INTEGERS

NAME _____ CLASS _____ DATE _____

DO NOW From memory. No notes.

- 1 How far is it from -5 to 3 ? 1.02
- 2 Start at -1 and move 6 left. Where do you land? 1.02
- 3 What is the opposite of -15 ? 1.01
- 4 Which integer is halfway between -6 and 2 ? 1.02

THINK FIRST Have a go before we learn it. It's fine to be unsure.

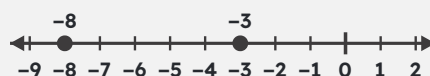
Which is colder: -8°C or -3°C ?

WORDS

$<$ means "is less than".
 $>$ means "is greater than".
ascending: smallest to largest.

BIG IDEA

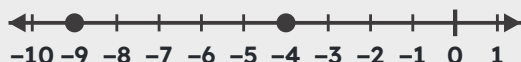
Further right means larger. So $-3 > -8$, even though 8 is more than 3.



LEARN IT Read each example. Then do the one beside it. Stuck? Look back at the example.

EXAMPLE 1

Which is larger: -9 or -4 ?



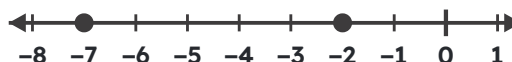
-4 is right of -9

larger: -4

Why? Sam says -9 is larger because 9 is more than 4. What would you tell Sam?

YOUR TURN 1

Which is larger: -2 or -7 ?



___ is right of ___

larger: ___

EXAMPLE 2

Write $<$ or $>$: -6 $\square$ -1

-6 is left of -1

$-6 < -1$

Why? How do you remember which way $<$ and $>$ point?

YOUR TURN 2

Write $<$ or $>$: 0 $\square$ -5

___ is right of ___

EXAMPLE 3

Order smallest to largest: $3, -5, 0, -2$

smallest: -5

$-5, -2, 0, 3$

Why? Where would your answer sit on a number line?

YOUR TURN 3

Order smallest to largest: $-1, 4, -6, 2$

CHECK Everyone answers. Circle one letter, then hold it up.

Which is the smallest?

- (A) -2 (B) 0 (C) -9 (D) -5

SPOT THE PATTERN

Write $<$ or $>$. Each question changes one thing.

1 $3 \square 5$

2 $-3 \square 5$

3 $-3 \square -5$

4 $3 \square -5$

5 $-5 \square -3$

6 $-30 \square -5$

7 $-5 \square 0$

8 $0 \square -30$

What do you notice? Compare 1 and 3. Putting a $-$ on both numbers did what to the sign?

MIXED PRACTICE

Today's skill mixed with earlier ones. Read each question carefully: they are not all the same kind.

1 Write $<$ or $>$: $-11 \square -12$

1.03

2 How far is it from -11 to -12 ?

1.02

3 Order largest to smallest: $-4, 7, -10, 0$

1.03

4 What is the opposite of -20 ?

1.01

5 Which is colder: -15°C or -12°C ?

1.03

6 Which integer is halfway between -12 and -4 ?

1.02

7 Write an integer that is between -5 and -3 .

1.03

8 Write $<$ or $>$: $-(-6) \square -6$

1.03

9 Start at -8 and move 11 right. Where do you land?

1.02

10 Order smallest to largest: $-30, -3, -13, 3$

1.03

GO FURTHER

Finished? Try these. Explain your thinking.

1 Always, sometimes or never? "A negative number is smaller than a positive number." Explain.

2 I am an odd integer. I am less than -2 and greater than -6 . What could I be?

3 Order these coldest first: $-7^{\circ}\text{C}, 4^{\circ}\text{C}, -12^{\circ}\text{C}, 0^{\circ}\text{C}$. How far apart are the coldest and warmest?

EXIT TICKET

On your own. Hand it in.

1 Write $<$ or $>$: $-8 \square -3$

2 Order smallest to largest: $2, -9, -1, -4$

3 Which is warmer: -6°C or -10°C ?

Today I can compare and order integers.

Not yet

Nearly

Got it

1.04 Adding integers

YEAR 7 · CHAPTER 1: INTEGERS

NAME _____ CLASS _____ DATE _____

DO NOW From memory. No notes.

1 $-7 \square -2$ Write $<$ or $>$.

1.03 2 Order smallest to largest: 4, -6, 0, -1

3 What is the opposite of -9?

1.01 4 Start at -4. Move 6 right. Where do you land? 1.02

THINK FIRST Have a go before we learn it. It's fine to be unsure.

It is -3°C at 6 am. By noon it is 5°C warmer. What is the temperature at noon?

WORDS

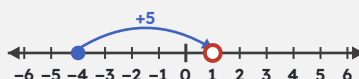
sum: the answer to an addition.

-3: say "negative three".

zero pair: +1 and -1 make 0.

BIG IDEA

Start at the first number. Adding a positive: move right. Adding a negative: move left.

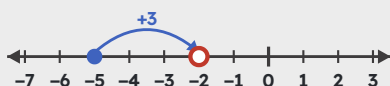


LEARN IT

Read each example. Then do the one beside it. Stuck? Look back at the example.

EXAMPLE 1

$$-5 + 3$$



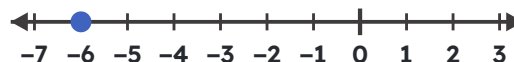
start at -5, move 3 right

$$= -2$$

Why? We moved right. Why?

YOUR TURN 1

$$-6 + 4$$

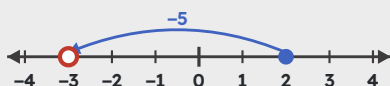


start at ____, move ____ right

$$= ______$$

EXAMPLE 2

$$2 + (-5)$$



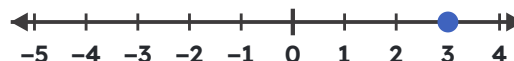
start at 2, move 5 left

$$= -3$$

Why? $2 - 5$ also moves 5 left from 2. What does that tell you about $2 + (-5)$?

YOUR TURN 2

$$3 + (-7)$$



start at ____, move ____ left

YOUR TURN 3

$$-2 + (-6)$$

EXAMPLE 3

$$-4 + (-3)$$

$$= -4 - 3$$

$$= -7$$

Why? Tom says $-4 + (-3) = 7$. What went wrong?

CHECK Everyone answers. Circle one letter, then hold it up.

Work out $-2 + 5$

(A) -7 (B) 3 (C) -3 (D) 7

SPOT THE PATTERN

Each question changes one thing. Work them out, then look down the answers.

1 $-3 + 3 =$ _____

2 $-3 + 2 =$ _____

3 $-3 + 1 =$ _____

4 $-3 + 0 =$ _____

5 $-3 + (-1) =$ _____

6 $-3 + (-2) =$ _____

7 $-3 + (-3) =$ _____

8 $-30 + (-30) =$ _____

What do you notice? What happens to the answer each time the number added goes down by 1?

MIXED PRACTICE

Today's skill mixed with earlier ones. Read each question carefully: they are not all the same kind.

1 $-8 + 5$

1.04

2 Which is larger: -8 or 5 ?

1.03

3 $6 + (-9)$

1.04

4 How far apart are -8 and 5 ?

1.02

5 $-7 + (-2)$

1.04

6 What is the opposite of -12 ?

1.01

7 $-15 + 15$

1.04

8 $-6 + \square = -1$

1.04

9 A diver is at -12 m. She rises 5 m. Where is she now? 1.04

10 Order smallest to largest: $-3 + 4$, -3 , $-3 + (-4)$ 1.03

GO FURTHER

Finished? Try these. Explain your thinking.

1 Always, sometimes or never true? "Adding a negative number makes the answer smaller." Explain.

2 Find two integers that add to -3 . Then find two more, where one of them is positive.

3 Start at -4 . Add 2 , 6 and -5 in any order. Where do you finish? Does the order matter?

EXIT TICKET

On your own. Hand it in.

1 $-7 + 3$

2 $4 + (-9)$

3 It is -6°C . It warms up 10°C . What is the temperature now?

Today I can add positive and negative integers.

Not yet

Nearly

Got it

1.05 Subtracting integers

YEAR 7 · CHAPTER 1: INTEGERS

NAME _____ CLASS _____ DATE _____

DO NOW From memory. No notes.

1 $-7 + 4$

1.04

2 $5 + (-8)$

1.04

3 Write $<$ or $>$: -6 $\square$ -2

1.03

4 What is the opposite of -5 ?

1.01

THINK FIRST Have a go before we learn it. It's fine to be unsure.

It is 3°C . It gets 5 degrees colder. What is the temperature now?

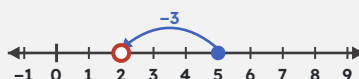
WORDS

difference: the answer to a subtraction.
 $5 - (-3)$: say "five subtract negative three".

opposite: swap the sign: 3 and -3 .

BIG IDEA

Subtracting is adding the opposite. $5 - 3 = 5 + (-3)$: move left. $5 - (-3) = 5 + 3$: move right.

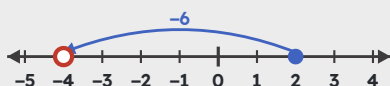


LEARN IT

Read each example. Then do the one beside it. Stuck? Look back at the example.

EXAMPLE 1

$2 - 6$



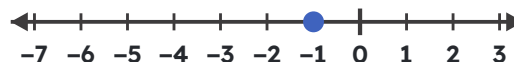
start at 2, move 6 left

$= -4$

Why? Why is $2 - 6$ negative?

YOUR TURN 1

$-1 - 4$

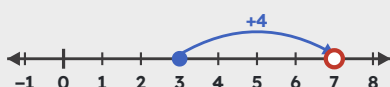


start at ____, move ____ left

$=$ ____

EXAMPLE 2

$3 - (-4)$



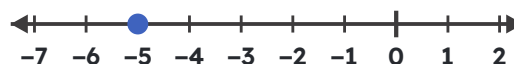
$= 3 + 4$

$= 7$

Why? Why does subtracting a negative move right?

YOUR TURN 2

$-5 - (-2)$



$=$ ____ $+$ ____

EXAMPLE 3

$-6 - (-9)$

$= -6 + 9$

$= 3$

Why? Ali says $3 - (-2) = 1$. What went wrong?

YOUR TURN 3

$-8 - (-3)$

CHECK Everyone answers. Circle one letter, then hold it up.

Work out $4 - (-3)$

- (A) 1 (B) 7 (C) -7 (D) -1

SPOT THE PATTERN

Each question subtracts 1 less than the one before.

1 $5 - 3 =$ _____

2 $5 - 2 =$ _____

3 $5 - 1 =$ _____

4 $5 - 0 =$ _____

5 $5 - (-1) =$ _____

6 $5 - (-2) =$ _____

7 $5 - (-3) =$ _____

8 $-5 - (-3) =$ _____

What do you notice? The number subtracted goes down by 1. What happens to the answer? What does that say about subtracting a negative?

MIXED PRACTICE

Today's skill mixed with earlier ones. Read each question carefully: they are not all the same kind.

1 $-3 - 6$

1.05

2 $-3 + 6$

1.04

3 $-3 - (-6)$

1.05

4 $-3 + (-6)$

1.04

5 $0 - 8$

1.05

6 Which is larger: $-3 - 6$ or $-3 + 6$?

1.03

7 It is -4°C . It gets 7°C colder. What is the temperature now?

1.05

8 How much warmer is 6°C than -2°C ? (warmer – colder)

1.05

9 How far is it from -9 to 6 ?

1.02

10 $10 - (-5) - 20$

1.05

GO FURTHER

Finished? Try these. Explain your thinking.

1 Is $7 - (-2)$ the same as $7 + 2$? Use a number line to explain.2 Find a positive integer and a second integer whose difference (first – second) is -5 .3 A submarine is at -120 m. A plane is at 350 m. How far apart are they?**EXIT TICKET**

On your own. Hand it in.

1 $-2 - 7$

2 $6 - (-5)$

3 $-4 - (-9)$

Today I can subtract positive and negative integers.

Not yet

Nearly

Got it

1.06 Multiplying integers

YEAR 7 · CHAPTER 1: INTEGERS

NAME _____ CLASS _____ DATE _____

DO NOW

From memory. No notes.

1 $-3 - 5$

1.05

2 $2 - (-6)$

1.05

3 $(-4) + (-4) + (-4)$

1.04

4 7×8

Year 6

THINK FIRST

Have a go before we learn it. It's fine to be unsure.

You lose \$3 a day for 4 days. By how much has your money changed?

WORDS

product: the answer to a multiplication.

same signs: both + or both -.

different signs: one + and one -.

BIG IDEA

Multiply the numbers, then choose the sign. Same signs: positive.

Different signs: negative.



LEARN IT

Read each example. Then do the one beside it. Stuck? Look back at the example.

EXAMPLE 1

$3 \times (-2)$



$(-2) + (-2) + (-2)$

$= -6$

Why? Why is $3 \times (-2)$ negative?

YOUR TURN 1

$4 \times (-3)$



$(___) + (___) + (___) + (___)$

$= ___$

EXAMPLE 2

-5×4

different signs: -

$5 \times 4 = 20$, so -20

Why? Is -5×4 the same as $4 \times (-5)$? Why?

YOUR TURN 2

$6 \times (-7)$

different signs: _____

EXAMPLE 3

$-3 \times (-6)$

same signs: +

$3 \times 6 = 18$, so 18

Why? Page 2 has a pattern that shows why $- \times -$ is positive. What does it show?

YOUR TURN 3

$-8 \times (-5)$

CHECK

Everyone answers. Circle one letter, then hold it up.

Work out $-4 \times (-5)$

(A) -20

(B) 20

(C) -9

(D) 9

SPOT THE PATTERN The first number goes down by 1 each time.

1 $3 \times (-4) =$

2 $2 \times (-4) =$

3 $1 \times (-4) =$

4 $0 \times (-4) =$

5 $-1 \times (-4) =$

6 $-2 \times (-4) =$

7 $-3 \times (-4) =$

8 $-3 \times 4 =$

What do you notice? What happens to the answer each time? What does that show about $- \times -$?

MIXED PRACTICE Today's skill mixed with earlier ones. Read each question carefully: they are not all the same kind.

1 -7×3

1.06

2 $-7 + 3$

1.04

3 $-7 \times (-3)$

1.06

4 $-7 - (-3)$

1.05

5 $9 \times (-6)$

1.06

6 $-2 \times 5 \times (-3)$

1.06

7 Which is larger: -2×6 or $-2 \times (-6)$?

1.03

8 A diver goes down 5 m every minute for 6 minutes. What is her change in height?

1.06

9 $\square \times (-4) = 20$

1.06

10 $8 - 15$

1.05

GO FURTHER Finished? Try these. Explain your thinking.

1 Find all the pairs of integers that multiply to -12 .

2 Always, sometimes or never? "Negative $\times$ negative $\times$ negative is negative." Explain.

3 Work out $(-1) \times (-1) \times (-1) \times (-1) \times (-1)$. What if there were 100 of them?

EXIT TICKET On your own. Hand it in.

1 -6×4

2 $-9 \times (-2)$

3 The temperature falls 3°C each hour for 5 hours. What is the total change?

Today I can multiply integers and choose the sign.

Not yet

Nearly

Got it

1.07 Dividing integers

YEAR 7 · CHAPTER 1: INTEGERS

NAME _____ CLASS _____ DATE _____

DO NOW

From memory. No notes.

1 -6×7

1.06

2 $-4 \times (-8)$

1.06

3 $-2 - 9$

1.05

4 Order smallest to largest: $-4, 2, -8$

1.03

THINK FIRST

Have a go before we learn it. It's fine to be unsure.

A debt of \$24 is shared equally by 4 people. How much does each person owe?

WORDS

quotient: the answer to a division.

fact family: $3 \times (-4) = -12$, so $-12 \div 3 = -4$.

share: split into equal groups.

BIG IDEA

Division undoes multiplication, so the sign rules are the same.

Same signs: positive. Different signs: negative.



LEARN IT

Read each example. Then do the one beside it. Stuck? Look back at the example.

EXAMPLE 1

$-12 \div 3$



$3 \times (-4) = -12$

so $-12 \div 3 = -4$

Why? How does the $\times$ fact help with the $\div$?

YOUR TURN 1

$-15 \div 5$



___ $\times$ (___) = ___

so ___ $\div$ ___ = ___

EXAMPLE 2

$28 \div (-4)$

different signs: -

$28 \div 4 = 7$, so -7

Why? How can you check that $28 \div (-4) = -7$?

YOUR TURN 2

$-36 \div 9$

different signs: _____

EXAMPLE 3

$-30 \div (-6)$

same signs: +

$30 \div 6 = 5$, so 5

Why? Jo says $-30 \div (-6) = -5$. What went wrong?

YOUR TURN 3

$-48 \div (-8)$

CHECK

Everyone answers. Circle one letter, then hold it up.

Work out $-20 \div (-4)$

(A) -5

(B) 5

(C) -16

(D) 80

SPOT THE PATTERN

Each question changes one sign or one number.

1 $12 \div 3 =$

2 $-12 \div 3 =$

3 $12 \div (-3) =$

4 $-12 \div (-3) =$

5 $-12 \div (-4) =$

6 $-12 \div (-6) =$

7 $-12 \div 6 =$

8 $-120 \div (-6) =$

What do you notice? Which answers are negative? What do their questions have in common?

MIXED PRACTICE

Today's skill mixed with earlier ones. Read each question carefully: they are not all the same kind.

1 $-42 \div 7$

1.07

2 -6×7

1.06

3 $54 \div (-9)$

1.07

4 $-9 - (-6)$

1.05

5 $-63 \div (-7)$

1.07

6 $-63 + 70$

1.04

7 $\square \div 4 = -5$

1.07

8 The temperature fell 18°C in 6 hours, the same each hour. What was the change each hour?

1.07

9 $-3 \times (-3) \times (-3)$

1.06

10 $-80 \div 4 \div (-2)$

1.07

GO FURTHER

Finished? Try these. Explain your thinking.

1 Find the average of -6 , 2 , -5 and -3 . (Add them, then divide by 4.)

2 Write three different divisions that have the answer -4 .

3 Always, sometimes or never? "Dividing makes a number smaller." Explain.

EXIT TICKET

On your own. Hand it in.

1 $-35 \div 5$

2 $-72 \div (-8)$

3 A debt of \$60 is shared by 3 people. Write each share as an integer.

Today I can divide integers and choose the sign.

Not yet

Nearly

Got it

1.08 Order of operations with integers

YEAR 7 · CHAPTER 1: INTEGERS

NAME _____ CLASS _____ DATE _____

DO NOW

From memory. No notes.

1 $-18 \div 3$

1.07

2 $-4 \times (-6)$

1.06

3 $5 - 9$

1.05

4 $-8 + 3$

1.04

THINK FIRST

Have a go before we learn it. It's fine to be unsure.

Two students work out $2 + 3 \times 4$. One gets 20. One gets 14. Who is right?

WORDS

order of operations: brackets, then $\times$ and $\div$, then $+$ and $-$.

left to right: for a tie, work from left to right.

evaluate: work out the value.

BIG IDEA

1. Brackets. 2. $\times$ and $\div$, left to right. 3. $+$ and $-$, left to right. The integer rules still work at every step.

LEARN IT

Read each example. Then do the one beside it. Stuck? Look back at the example.

EXAMPLE 1

$-3 + 4 \times 2$

$= -3 + 8$

$= 5$

Why? Why do we do 4×2 first?

YOUR TURN 1

$-10 + 3 \times 2$

$= ___ + ___$

$= ___$

EXAMPLE 2

$(-3 + 7) \times 2$

$= 4 \times 2$

$= 8$

Why? How did the brackets change the answer?

YOUR TURN 2

$(1 - 6) \times 3$

$= ___ \times ___$

EXAMPLE 3

$6 - 2 \times (-3)$

$= 6 - (-6)$

$= 6 + 6$

$= 12$

Why? Mia says $6 - 2 \times (-3) = -12$. What went wrong?

YOUR TURN 3

$-5 - 4 \times (-2)$

CHECK

Everyone answers. Circle one letter, then hold it up.

Work out $10 - 3 \times (-2)$

(A) -14

(B) 4

(C) 16

(D) -16

SPOT THE PATTERN The same numbers, with one change each time.

1 $2 + 3 \times 4 =$ _____

2 $2 - 3 \times 4 =$ _____

3 $-2 - 3 \times 4 =$ _____

4 $-2 - 3 \times (-4) =$ _____

5 $(-2 - 3) \times (-4) =$ _____

6 $(-2 + 3) \times (-4) =$ _____

7 $-2 + 3 \times (-4) =$ _____

8 $-20 + 30 \times (-4) =$ _____

What do you notice? Find two questions with the same numbers but different answers. What made them different?

MIXED PRACTICE Today's skill mixed with earlier ones. Read each question carefully: they are not all the same kind.

1 $-4 + 5 \times 2$

1.08

2 $(-4 + 5) \times 2$

1.08

3 $-12 \div 4 - 3$

1.08

4 $-12 \div (-4)$

1.07

5 $8 - 2 \times 7$

1.08

6 $-3 \times (2 - 6)$

1.08

7 $-5 \times (-5)$

1.06

8 $20 \div (-5) + (-6)$

1.08

9 $-2 \times 3 + 4 \times (-2)$

1.08

10 $(-9 + 3) \div (-2)$

1.08

GO FURTHER Finished? Try these. Explain your thinking.

1 Put brackets in to make it true: $-4 + 6 \times 3 = 6$

2 Use -2 , 3 and 5 once each, with $+$ and $\times$, and brackets if you like. What is the smallest answer you can make?

3 Write a calculation with brackets, $\times$ and $-$ that has the answer -12 .

EXIT TICKET On your own. Hand it in.

1 $-6 + 2 \times 4$

2 $(-6 + 2) \times 4$

3 $9 - 3 \times (-2)$

Today I can use the order of operations with integers.

Not yet

Nearly

Got it

1.09 Integers on a calculator

YEAR 7 · CHAPTER 1: INTEGERS

NAME _____ CLASS _____ DATE _____

DO NOW

From memory. No notes.

1 $-3 + 2 \times 5$

1.08

2 $(4 - 9) \times 2$

1.08

3 $-56 \div 8$

1.07

4 $-7 \times (-6)$

1.06

THINK FIRST

Have a go before we learn it. It's fine to be unsure.

Without a calculator: is -48×21 about -100 , about $-1\ 000$, or about $-10\ 000$?

WORDS

$(-)$ key: makes a number negative.

$-$ key: subtracts.

estimate: a quick answer from rounded numbers.

BIG IDEA

Use $(-)$ for a negative number, $-$ to subtract. Got a $+/-$ key? Press it after the number. Estimate first, then check.

$(-)$ 4 8 $\times$ 2 1 =

LEARN IT

Read each example. Then do the one beside it. Stuck? Look back at the example.

EXAMPLE 1

$-37 + 85$

keys: $(-)$ 37 + 85 =

= 48

Why? What could go wrong if you pressed $-$ instead of $(-)$ at the start?

YOUR TURN 1

$-126 + 59$

keys: $(-)$ ___ + ___ =

= ___

EXAMPLE 2

-48×21

estimate: $-50 \times 20 = -1\ 000$

= $-1\ 008$

Why? How does the estimate help?

YOUR TURN 2

-39×31

estimate: ___ $\times$ ___ = ___

EXAMPLE 3

$(-35 - 47) \times 12$

bracket: -82

= -984

Why? Why must you type the brackets?

YOUR TURN 3

$(-28 + 13) \times (-6)$

CHECK

Everyone answers. Circle one letter, then hold it up.

Ella presses $(-) 6 - 4 =$ What does her calculator show?

(A) -2

(B) 2

(C) -10

(D) 10

SPOT THE PATTERN

Use your calculator. Each question changes one thing.

1 $-25 \times 4 =$

2 $-25 \times 8 =$

3 $-25 \times 12 =$

4 $-25 \times 16 =$

5 $-25 \times (-16) =$

6 $25 \times (-16) =$

7 $-250 \times 16 =$

8 $-250 \times (-160) =$

What do you notice? What happens when one number gets bigger by 4? When one sign changes?

MIXED PRACTICE

Today's skill mixed with earlier ones. Read each question carefully: they are not all the same kind.

1 $-347 + 128$

1.09

2 -56×23

1.09

3 $-1\,248 \div (-24)$

1.09

4 No calculator: $-6 + 3 \times (-2)$

1.08

5 $(-64 - 36) \times 15$

1.09

6 Estimate -62×19 . Then use your calculator.

1.09

7 No calculator: $-8 \times (-9)$

1.06

8 A submarine at -185 m rises 12 m a minute for 9 minutes. Where is it now?

1.09

9 $-2\,016 \div 42$

1.09

10 Order smallest to largest: $-125, -152, -98$

1.03

GO FURTHER

Finished? Try these. Explain your thinking.

1 Use your calculator to find the missing number: $-37 \times \square = 1\,147$

2 A student gets 1 008 for -48×21 . How can you tell it is wrong without working it out?

3 Moscow had these temperatures: $-12, -8, -15, -3$ and -7°C . Find the average.

EXIT TICKET

On your own. Hand it in.

1 $-234 + 97$

2 $-38 \times (-27)$

3 $(-45 - 27) \div 8$

Today I can use a calculator with integers and check answers.

Not yet

Nearly

Got it

1.10 Integer problems

YEAR 7 · CHAPTER 1: INTEGERS

NAME _____ CLASS _____ DATE _____

DO NOW

From memory. No notes.

1 $-144 \div 12$

1.07

2 $-2 \times (3 - 7)$

1.08

3 $-5 - (-8)$

1.05

4 Which is warmer: 1.03

-14°C or -9°C ?

THINK FIRST

Have a go before we learn it. It's fine to be unsure.

A lift at level -2 goes up 6 floors, then down 3 floors. Which level is it on now?

WORDS

balance: how much money is in an account.

overdrawn: the balance is below \$0.

change: new $-$ old.

BIG IDEA

Read the story. Write a number sentence. Up, rise, earn: $+$. Down, fall, spend: $-$. A change that repeats: $\times$.

LEARN IT

Read each example. Then do the one beside it. Stuck? Look back at the example.

EXAMPLE 1

It is -4°C at 6 am. By noon it is 9°C warmer. What is the temperature at noon?

$-4 + 9$

$= 5^{\circ}\text{C}$

Why? How did you choose $+$ or $-$?

YOUR TURN 1

It is 3°C at 6 pm. By midnight it is 8°C colder. What is the temperature at midnight?

$___ - ___$

$= ___^{\circ}\text{C}$

EXAMPLE 2

Sam has \$45. Sam spends \$70. What is the balance?

$45 - 70$

$= -\$25$

Why? What does a balance of $-\$25$ mean?

YOUR TURN 2

Ana has $-\$30$. She is paid \$55. What is her balance?

$___ + ___$

EXAMPLE 3

A diver at -10 m goes down 3 m a minute for 4 minutes. Where is she now?

$4 \times (-3) = -12$

$-10 + (-12) = -22$ m

Why? Why multiply first?

YOUR TURN 3

A diver at -40 m rises 5 m a minute for 6 minutes. Where is she now?

CHECK

Everyone answers. Circle one letter, then hold it up.

A gull is at 15 m. A shark is at -20 m. How far apart are they?

(A) 5 m

(B) 35 m

(C) -5 m

(D) -35 m

SPOT THE PATTERN

All four operations with -5 . Each changes one thing.

1 $-5 + 8 =$ _____

2 $-5 + 5 =$ _____

3 $-5 + 2 =$ _____

4 $-5 - 2 =$ _____

5 $-5 - (-2) =$ _____

6 $-5 \times 2 =$ _____

7 $-5 \times (-2) =$ _____

8 $-5 \div (-5) =$ _____

What do you notice? Which two questions have the same answer? Why?

MIXED PRACTICE

Today's skill mixed with earlier ones. Read each question carefully: they are not all the same kind.

1 It is -8°C . It rises 12°C . What is the temperature now?

1.10

2 How many degrees warmer is 6°C than -9°C ?

1.10

3 A lift at level 3 goes down 7 floors. Which level is it on?

1.10

4 A balance is \$20. You spend \$35, then earn \$10. Balance?

1.10

5 $-4 \times 3 + 20$

1.08

6 A plane is at 900 m. A submarine is at -150 m. How far apart are they?

1.10

7 The temperature falls 4°C each hour for 6 hours. What is the total change?

1.10

8 A debt of \$90 is shared by 3 people. Write each share as an integer.

1.10

9 A quiz: +3 if right, -1 if wrong. Kai gets 6 right, 5 wrong. His score?

1.10

10 Order coldest first: -3°C , -13°C , 3°C , 0°C

1.03

GO FURTHER

Finished? Try these. Explain your thinking.

1 In the quiz, Lea answered 10 questions and scored 14. How many were right?

2 It is -6°C . It warms 2°C an hour for 12 hours, then cools 3°C an hour for 4 hours. Temperature?

EXIT TICKET On your own. Hand it in.

1 It is -7°C . It rises 11°C . What is the temperature now?

2 A balance is \$15. You spend \$40. What is the balance?

3 A diver at -8 m goes down 2 m a minute for 5 minutes. Where is she?

Today I can solve problems with integers.

Not yet

Nearly

Got it

NAME _____ CLASS _____ DATE _____

MIXED REVIEW A

No notes. Show your working. Calculator only where it says so.

1 What is the opposite of -17 ?

1.01

2 $-(-8)$

1.01

3 How far is it from -6 to 9 ?

1.02

4 Which integer is halfway between -14 and 2 ?

1.02

5 Write $<$ or $>$: -12 $\square$ -21

1.03

6 Order smallest to largest: $5, -8, 0, -3, -10$

1.03

7 $-9 + 14$

1.04

8 $3 + (-11)$

1.04

9 $-4 - 7$

1.05

10 $-2 - (-10)$

1.05

11 -7×8

1.06

12 $-6 \times (-9)$

1.06

MIXED REVIEW B

Keep going. Each question says which lesson it comes from.

13 $-81 \div 9$

1.07

14 $-64 \div (-8)$

1.07

15 $-5 + 3 \times (-4)$

1.08

16 $(-5 + 3) \times (-4)$

1.08

17 Calculator: $-384 \div 16$

1.09

18 Calculator: $(-73 + 28) \times 11$

1.09

19 It is -3°C . It falls 9°C . What is the temperature now?

1.10

20 A diver at -25 m rises 4 m a minute for 5 minutes. Where is she now?

1.10

21 How far apart are 40 m and -35 m?

1.10

22 $-2 \times (-3) \times (-5)$

1.06

23 Is $6 - (-4)$ the same as $6 + 4$? Explain.

1.05

24 Put brackets in to make it true: $2 - 8 \times 3 = -18$

1.08

WHAT NEXT?

Mark your answers with your teacher. Tick each lesson where you got every question right.

1.01 ☐ 1.02 ☐ 1.03 ☐ 1.04 ☐ 1.05 ☐ 1.06 ☐ 1.07 ☐ 1.08 ☐ 1.09 ☐ 1.10 ☐

Not ticked? Go back to that lesson's Examples, then try its Exit ticket again.